Validation did not confirm development-only selections

Report 2 of 3 · Validation and frozen selection

Per-coin P&L and joint basket alpha/betas, with confidence intervals, for all nine frozen strategies.

Executive Summary

The main finding is failure to generalize across periods

Follow-up orientation through 30 September 2026, not an input to the original development selection. All four development-selected Track A strategies lost in 2024-2025; their raw Sharpes ranged from **-0.64 to -0.22 (365-day)**, or **-0.53 to -0.18 (252-day)**. All five validation-informed Track B strategies lost compounded net return in January-September 2026. B06 still had slightly positive arithmetic raw Sharpe, **0.05 / 0.04 (365 / 252)**; a compounded loss does not require negative Sharpe.

This pattern is **consistent with overfitting or weak transfer, not proof of one cause**: baskets, settings and cadence differ, and Track B also adds long/cash trend. A03's **+31.38%** in 2026 and raw Sharpe **2.32 / 1.93** do not erase its **-22.31%** validation loss. No new winner or reliable all-period strategy is established.

Sharpe convention: tables show **365 / 252** on the **same UTC daily series including weekends**. The alternative is S252 = S365 × √(252/365) = 0.8309097 × S365; it does not drop weekends. An ≈ marks a 252-day value available only from a rounded 365-day source. The primary 365-day results, inference, rankings, CAGR and alpha are unchanged. Full-precision convention comparison.

Separate fixed all-22 studies did not establish persistence. Development / validation / Jan-Sep 2026 net returns were **+5.61% / -17.43% / -20.94%** for momentum and **-33.59% / -15.66% / +9.91%** for reversal. Neither produced positive net returns in all three periods. These remain separate fixed studies, not a new direction selection or fresh independent confirmation. Study results, separate tests and chronology disclosures.

Validation findings

- **All four development-selected candidates lost in 2024–2025**, with net returns from **–22.31% to –45.31%** at 20 bps. Their positive development Sharpes did not generalize to the next two years.
- The broad development-Q1 group retained a useful relative ordering, but not positive median economics: its validation median return was **–16.53%** and **30.92%** of cases were profitable. “Q1” preserves the original best-approximately-20% development-rank labels within each basket, not a calendar quarter; current counts are retained subsets, not newly ranked quintiles.
- A separate validation-informed rule produced **five Track B candidates** from **39,332 eligible pairs**. Their validation Sharpes of **1.14–1.85 (365-day), or 0.95–1.54 (252-day), are selection statistics**, not independent evidence of discovery.
- The final research shortlist preserves the **original four plus five Track B selections**. No single combined portfolio, weights or winner is implied.

Read the periods correctly

Development: 1 January 2022–31 December 2023. Design and diagnose the rule using this period only.
Validation: 1 January 2024–31 December 2025. Test Track A unchanged, then explicitly label later validation-informed selection as Track B.

The reserved evaluation was subsequently approved for **1 January–30 September 2026**, with an October 1 UTC closing endpoint. Detailed outcomes belong in Report 3. Performance and factor diagnostics use the audited saved strategy paths and existing factor prices; signals, trades and selections are fixed.

Primary convention: 20 bps one-way fees on full-L1 target changes. Lag1 means idealized observed-close execution, not a real one-hour delay.

Separate fixed all-22 studies did not establish persistent performance

These are **two separately approved simplicity studies**, responding to the proposal to remove basket/parameter selection. Each fixes one 28-day cross-sectional rank-demean rule on all 22 eligible economic slots, with Monday 00:00 UTC rebalancing and no buffer. Momentum buys relative winners and shorts losers; the independent reversal study buys relative losers and shorts winners. No direction is selected, no tests are pooled, and neither changes the nine frozen strategies.

Momentum gained modestly in development, then lost in validation and partial-year 2026. **Reversal** lost in development and validation; its positive partial 2026 does not establish persistence. Neither produced positive net returns in all three periods. The pattern across periods—not the best isolated result—is the finding.

Standalone momentum: unchanged 20 bps / lag1 primary

Period	Net return	Net Sharpe 365 / 252	Hourly max DD	Annual full-L1 turnover
2022–2023 development	5.61%	0.23 / 0.20	29.53%	33.58×
2024–2025 validation	-17.43%	-0.38 / -0.31	26.56%	34.91×
Jan–Sep 2026 (273 days)	-20.94%	-1.70 / -1.41	22.06%	39.29×

Standalone reversal: unchanged 20 bps / lag1 primary

Period	Net return	Net Sharpe 365 / 252	Hourly max DD	Annual full-L1 turnover
2022–2023 development	-33.59%	-0.87 / -0.72	43.69%	33.58×
2024–2025 validation	-15.66%	-0.32 / -0.27	31.57%	34.91×
Jan–Sep 2026 (273 days)	9.91%	0.81 / 0.67	11.10%	39.29×

2024–2025 validation: additional primary risk and growth

Study	Gross return	CAGR estimate	Net volatility √365	Daily max DD	Hourly DD episode
Momentum	-5.02%	-9.12%	20.06%	25.48%	Peak 2024-01-02 Trough 2025-01-06 Not recovered; censored 729.7 days underwater
Reversal	-2.99%	-8.15%	20.23%	30.85%	Peak 2024-03-02 Trough 2025-11-03 Not recovered; censored 669.8 days underwater

Measurement. Net returns include transaction fees; gross returns exclude them. Both Sharpe annualization conventions use the same UTC daily observations including weekends: $S252 = S365 \times \sqrt{(252/365)}$. CAGR, volatility and alpha use 365 days. Daily drawdown and original hourly drawdown use different endpoint clocks; both include initial NAV. Turnover is full-L1 traded-weight multiples including entry, forced events and liquidation. Each two-year return is compounded; CAGR is annualized.

2024–2025 validation: uncertainty remains material

The saved **net daily-mean HAC27 t-statistics are descriptive**. Net Sharpe intervals below are **marginal and conditional** on the fixed rule and historical sample, not simultaneous or selection-adjusted evidence for all prior research. No new hypothesis tests or strategy selections are performed.

2024–2025 validation: descriptive net-return uncertainty

Study	Net-mean HAC27 t	Net Sharpe 95% CI: 365 / 252
Momentum	-0.694	365 [-1.38, 0.68] 252 [-1.14, 0.57]
Reversal	-0.598	365 [-1.41, 0.69] 252 [-1.17, 0.57]

2024–2025 validation: descriptive BTC sensitivity, marginal HAC7 intervals

Study	Alpha/year [95% CI]	BTC beta [95% CI]
Momentum	-4.38% [-28.81%, 20.05%]	-0.066 [-0.113, -0.019]
Reversal	-9.54% [-34.00%, 14.92%]	0.064 [0.017, 0.112]

The cross-period result is weaker than an isolated profitable window

Neither standalone rule was profitable after costs in all three periods. Momentum's hourly drawdowns stayed below the fixed 40% guardrail; reversal's development hourly drawdown reached **43.69%**. These observations do not support a deployment-ready, persistent edge.

Uncertainty methods: Bartlett HAC27 covariance with n/(n–1) for the descriptive mean t-statistic; 10,000 circular 28-day bootstrap draws per period, PCG64 seed 20261020 reset per period. Net Sharpe intervals are percentile intervals from the saved draws; the 252-day endpoints are the same positive rescaling of the 365-day endpoints. All draws were valid. Fees/trades are not reinserted at resampled block boundaries.

BTC coefficients: alpha is 365 times the daily arithmetic intercept, not CAGR. Full-calendar HAC7 intervals are descriptive. These are BTC-only estimates, **not joint 22-coin regressions**, and are not selection gates.

Preserve the procedural distinction

Momentum: its code/settings seal preceded the first all-22 simulation, but existing verification tests had already decoded packaged panels and recalculated saved-nine derived statistics. Strict pre-panel-access compliance is **false**; the chronology exception and corrections remain in its evidence. **Reversal:** its own freeze preceded the first new-study panel decode; strict pre-access checks passed.

All three historical windows had already been examined in the wider project. These additional studies **are not fresh independent OOS confirmation**. A failed rule here does not prove all momentum or reversal strategies fail. No basket, lookback, cadence, direction or fee sensitivity is promoted after seeing these outcomes.

2024–2025 validation: all predeclared sensitivities retained

Primary remains **20 bps / lag1**. Lag1 is idealized observed-close execution, not a guaranteed one-hour wait; lag2 adds one hour. Each independent study completed **21** scenario-period keys, with no missing or failed slots. These tables retain all seven scenarios for this report's period; no sensitivity becomes a new selection.

Standalone momentum: 2024–2025 validation

Cost bps / lag	Net return	Net Sharpe 365 / 252	Hourly max DD
0 / 1	-5.02%	-0.03 / -0.02	20.83%
5 / 1	-8.29%	-0.11 / -0.10	22.30%
10 / 1	-11.44%	-0.20 / -0.17	23.75%
15 / 1	-14.48%	-0.29 / -0.24	25.17%
20 / 1	-17.43%	-0.38 / -0.31	26.56%
40 / 1	-28.22%	-0.72 / -0.60	35.02%
20 / 2	-17.51%	-0.38 / -0.32	26.66%

Standalone reversal: 2024–2025 validation

Cost bps / lag	Net return	Net Sharpe 365 / 252	Hourly max DD
0 / 1	-2.99%	0.03 / 0.02	25.16%
5 / 1	-6.33%	-0.06 / -0.05	26.08%
10 / 1	-9.55%	-0.15 / -0.12	27.47%
15 / 1	-12.66%	-0.23 / -0.19	29.55%
20 / 1	-15.66%	-0.32 / -0.27	31.57%
40 / 1	-26.69%	-0.66 / -0.55	39.08%
20 / 2	-15.76%	-0.32 / -0.27	32.04%

Full universe, data and supporting evidence

Fixed historical-liquidity slots: BTC, ETH, SHIB, BNB, SOL, XRP, DOT, SAND, legacy LUNA/LUNC, AVAX, DOGE, ADA, FTM/S, TRX, MATIC/POL, MANA, FIL, GALA, LINK, AXS, LTC and VET. Eligible members are reranked together; BTC/ETH are ordinary members, not fixed sleeves. No reconstitution. Inherited inventory/survivor limitations remain.

Source: Binance Data Vision spot-USDT hourly archives; **43,824** normalized endpoints from **1 October 2021 01:00 UTC through 1 October 2026 00:00 UTC**, including warm-up, with seven pinned panels and explicit lifecycle/halt masks. No data were downloaded, gaps filled or windows shortened. Full data manifest. Shorts are theoretical; borrow, funding, margin, impact, capacity, realistic fills and ordinary drift-maintenance are unmodelled. USDT/USD parity is assumed.

Complete three-period evidence: Momentum results, scenarios, uncertainty. Reversal results, scenarios, uncertainty.

The nine frozen strategies: validation performance

2024–2025; primary 20 bps costs, all nine fixed IDs. This nine-case comparison is separate from the full-grid family distributions; it is not a new selection. Track A tests the development-selected procedure. Track B was selected using development plus validation: its validation Sharpe is a selection statistic, not an independent final-test estimate.

All nine fixed strategies: net performance

ID	Net return	CAGR estimate	Net Sharpe 365 / 252	Annual full-L1 turnover	Hourly max DD
A01	-29.55%	-16.05%	-0.39 / -0.33	35.15×	44.82%
A02	-39.63%	-22.28%	-0.64 / -0.53	55.43×	47.29%
A03	-22.31%	-11.84%	-0.39 / -0.33	22.34×	43.11%
A04	-45.31%	-26.02%	-0.22 / -0.18	63.91×	70.96%
B01	+120.19%	48.31%	1.62 / 1.35	77.45×	15.50%
B03	+151.30%	58.43%	1.85 / 1.53	24.13×	22.08%
B04	+73.78%	31.78%	1.14 / 0.95	22.34×	23.42%
B05	+152.00%	58.65%	1.40 / 1.17	12.48×	32.48%
B06	+164.24%	62.45%	1.24 / 1.03	24.59×	32.83%

Net results include 20 bps transaction costs. Returns are full-period compounded changes; CAGR and Sharpe are annualized estimates. Both Sharpe conventions use the same UTC daily returns including weekends: S252 = S365 × $\sqrt{(252/365)}$. Hourly maximum drawdown and full-L1 turnover retain the original accounting clock. BTC alpha is an arithmetic regression intercept annualized with 365, not CAGR.

1. Track A tested development robustness, not the validation winner

The original development-only screen required positive 2022 and 2023 returns and Sharpes, positive full-development returns under 40 bps and an additional-hour execution delay, and more than half of both parameter and all 28 one-swap basket neighbors to be profitable.

The robustness score was the minimum of the two annual Sharpes and the two neighborhood median Sharpes. At most one case-basket was selected per family; score ties used lower turnover, experiment ID and basket ID. No family gate was relaxed when empty.

The rule was designed after development robustness had been inspected. Freezing it prevents later reselection; it does not remove development search bias or earlier historical inspection.

The same four identities: net results in both periods

2022–2023 development

ID	Net return	CAGR estimate	Net Sharpe 365 / 252	Annual full-L1 turnover	Hourly max DD
A01	+165.36%	62.90%	1.84 / 1.53	31.44×	23.25%
A02	+252.94%	87.87%	2.03 / 1.69	49.58×	58.37%
A03	+64.38%	28.21%	1.19 / 0.99	27.62×	19.79%
A04	+314.22%	103.52%	1.43 / 1.19	50.50×	51.35%

2024–2025 validation

ID	Net return	CAGR estimate	Net Sharpe 365 / 252	Annual full-L1 turnover	Hourly max DD
A01	-29.55%	-16.05%	-0.39 / -0.33	35.15×	44.82%
A02	-39.63%	-22.28%	-0.64 / -0.53	55.43×	47.29%
A03	-22.31%	-11.84%	-0.39 / -0.33	22.34×	43.11%
A04	-45.31%	-26.02%	-0.22 / -0.18	63.91×	70.96%

Returns are compounded net of 20 bps transaction costs. Sharpe uses both annualization conventions, 365 / 252, on the same daily series. Turnover is annual full-L1 target change, including entry and liquidation. Hourly max DD is the positive loss magnitude.

All four failed the economic expectation of positive net validation returns. The result is not simply that they failed to remain rank one. Original risk also persisted: A04's validation maximum drawdown reached **70.96%**. The four stay in the record unchanged rather than being replaced by favorable controls.

2. The reference controls isolate a basket comparison

“Reference” means the **predefined liquidity-tier reference basket**, ID 0: BTC, ETH, SHIB, BNB, AVAX, DOGE, FIL and GALA. It is a fixed comparison, not a claim that this basket is optimal.

Each row compares exactly the same strategy settings on the selected basket and reference basket. The selected baskets had much stronger development results. In validation, reference momentum and long/short trend were slightly profitable, while their selected-basket versions lost. Reversal was worse on the reference basket, so there is no universal reference advantage.

Same settings, selected versus reference basket

ID	Period	Selected net	Reference net	Selected Sharpe (365 / 252)	Reference Sharpe (365 / 252)
A01	Development	+165.36%	+8.19%	1.84 / 1.53	0.28 / 0.23
A01	Validation	-29.55%	-9.90%	-0.39 / -0.33	-0.06 / -0.05
A02	Development	+252.94%	-13.26%	2.03 / 1.69	-0.10 / -0.08
A02	Validation	-39.63%	+8.66%	-0.64 / -0.53	0.29 / 0.24
A03	Development	+64.38%	+12.63%	1.19 / 0.99	0.35 / 0.29
A03	Validation	-22.31%	-38.60%	-0.39 / -0.33	-0.73 / -0.61
A04	Development	+314.22%	+98.46%	1.43 / 1.19	0.86 / 0.72
A04	Validation	-45.31%	+5.45%	-0.22 / -0.18	0.35 / 0.29

A basket substitution changes every cross-sectional rank and hence other coins' positions; trend models retain fixed per-slot budgets. This is a matched strategy comparison, not a causal coin-removal experiment. Validation-profitability of a control is evidence to retain and investigate, not permission to silently promote it.

3. The basket comparison also changes risk

The same-pair comparison must retain turnover and drawdown, not just returns and Sharpe. These rows use the identical strategy settings, dates, 20 bps fee convention and idealized observed-close execution as the preceding table.

Annual TO is full-L1 target-change turnover in times per year. Maximum drawdown is the positive loss magnitude from the saved continuous two-year path. This table is a comparison of realized paths, not a risk-adjusted portfolio or a causal basket treatment.

Reference risk was not uniformly lower. A03 reversal had a larger validation drawdown on the reference basket (**53.46%**) than on the selected basket (**43.11%**), despite slightly lower turnover. A04's reference drawdown was lower, but still **58.37%**.

Same settings, selected versus reference risk

ID	Period	Selected annual TO	Reference annual TO	Selected max DD	Reference max DD
A01	Development	31.44×	32.69×	23.25%	35.33%
A01	Validation	35.15×	33.52×	44.82%	30.83%
A02	Development	49.58×	52.75×	58.37%	44.13%
A02	Validation	55.43×	50.74×	47.29%	22.08%
A03	Development	27.62×	24.00×	19.79%	32.31%
A03	Validation	22.34×	21.41×	43.11%	53.46%
A04	Development	50.50×	54.25×	51.35%	56.16%
A04	Validation	63.91×	65.29×	70.96%	58.37%

4. Strong development paths depended on period-specific positions

The four candidates were profitable in both development years by construction. Validation exposed a different pattern: A01 recovered in 2025 after a deep 2024 loss; A02–A04 lost in both validation years. Two-year windows and annual views answer different questions, so they are shown together rather than moving window boundaries after seeing results.

Annual net returns on continuous paths

ID	2022	2023	2024	2025
A01	+66.26%	+59.60%	-39.15%	+15.77%
A02	+140.95%	+46.47%	-32.43%	-10.66%
A03	+32.05%	+24.48%	-1.91%	-20.80%
A04	+83.54%	+125.69%	-13.10%	-37.06%

Legacy LUNA slot: saved net-P&L attribution

ID	Development LUNA contribution (pp)	Validation LUNA contribution (pp)
A01	+49.39	-25.46
A02	+132.19	-19.51
A04	+78.96	-19.88

The legacy LUNA slot contributed **+49.39, +132.19 and +78.96 percentage points** to A01, A02 and A04's development net returns, respectively, but negative contributions in validation. These are NAV-weighted accounting contributions to starting capital, not standalone coin returns. The table reports percentage-point (pp) contributions.

All four selected baskets also lost **before modeled fees** in validation: gross compounded returns ranged from **−15.03% to −29.34%**. Costs deepened the losses rather than explaining them in full.

The diagnostic supports **concentration and period-dependence concerns**. It does not establish that deleting LUNA would have produced the same remaining positions or returns. Documented halts, forced exits and symbol lifecycle treatment were preserved; no event was removed to improve the backtest.

5. Relative rank transfer was stronger than economic transfer

The full grid retained **1,111,320 basket–configuration pairs**: 6,615 baskets × 168 retained settings. Original development-ranked memberships were frozen before full-grid validation and remain unchanged in this retained subset; the 10 accepted case-baskets retain their audited saved paths.

Q1 retained a higher validation median Sharpe than lower-ranked groups, but its median net return remained negative. Much of the rank hierarchy reflects the persistent cost burden of fast settings, not proof that a positive-alpha winner transfers.

These medians are constituent distributions, **not portfolios**. All retained development cases have defined Sharpe. No value is zero-filled, and no retained case is reassigned to a different original development group.

Frozen development-ranked groups

2022–2023 development

Group	Cases	Median net	Median Sharpe (365 / 252)	Median annual TO	Positive fraction
Q1 (best)	223512	-1.72%	0.14 / 0.11	60.81×	47.14%
Q2	219923	-37.53%	-0.74 / -0.61	70.69×	0.08%
Q3	219065	-69.08%	-1.77 / -1.47	159.47×	0.00%
Q4	223910	-92.90%	-4.03 / -3.35	487.59×	0.00%
Q5 (worst)	224910	-99.9973%	-15.69 / -13.04	2753.01×	0.00%

2024–2025 validation

Group	Cases	Median net	Median Sharpe (365 / 252)	Median annual TO	Positive fraction
Q1 (best)	223512	-16.53%	-0.14 / -0.12	64.93×	30.92%
Q2	219923	-36.18%	-0.75 / -0.62	70.05×	13.23%
Q3	219065	-59.58%	-1.63 / -1.35	162.09×	2.68%
Q4	223910	-90.15%	-4.18 / -3.48	489.30×	0.01%
Q5 (worst)	224910	-99.9977%	-17.87 / -14.85	2654.28×	0.00%

6. A high rank need not imply a positive return

Across the full grid, **62.71% of strictly positive retained development results became strictly negative** in validation. Conversely, **6.43% of strictly negative retained development results became positive**. The denominators differ, so these are conditional transition rates rather than symmetric probabilities. Exact zeros remain separate.

The retained subset of the original best development-ranked group has **223,512 cases**; **69,108 (30.92%)** were profitable in validation. Of these, **39,332** also had positive development returns. The remaining **29,776** were development-negative cases that survived validation. These counts are not independent replications.

How often the return sign changed

Family	Dev positive	Positive → negative	Dev negative	Negative → positive
All families	105552	62.71%	1005768	6.43%
Momentum	49083	72.29%	466887	8.69%
Reversal	13497	90.91%	502473	1.79%
Buffered momentum	10719	66.24%	15741	35.04%
Long/cash trend	14898	18.53%	11562	63.81%
Long/short trend	17355	49.43%	9105	24.91%

Development–validation Sharpe rank correlation

Comparison	Cohorts	Median Spearman	10th percentile	90th percentile
Across settings within basket	6615	0.91	0.86	0.94
Across baskets within setting	168	0.14	-0.23	0.62

The median rank correlation was **0.91 across settings within a basket**, but only **0.14 across baskets within the same setting**. That distinction explains why broad setting hierarchies can persist while the exact development-winning basket is unstable.

No IID p-value or confidence interval is attached to these full-grid cross-sectional summaries. Adjacent baskets share seven coins; configurations share signals, prices and dates. Long/cash trend's lower sign-flip rate is descriptive, not a new final-test endorsement.

7. Track B was an honest validation-informed selection

After validation had been inspected, a separate rule was frozen **before executing its selection**. Start with development Q1, require evaluated statuses, finite defined Sharpes and strictly positive net returns in both development and validation at 20 bps.

Within each of the five retained families, select the highest validation Sharpe. Ties use lower validation full-L1 turnover, lexical experiment ID and numeric basket ID. Select none when no case qualifies. No positive-Sharpe threshold, annual consistency, stress-cost survival, neighborhood success or risk cap was added as a hidden eligibility gate.

All five selected validation Sharpes are selection statistics from a large correlated pool. They are not unbiased final-test estimates and do not retroactively repair Track A.

Eligibility within the five retained families

ID	Family	Full grid	Dev Q1	Eligible	Selected
B01	Buffered momentum	26460	20068	3619	1
B03	Momentum	515970	107412	13585	1
B04	Reversal	515970	45673	1227	1
B05	Long/cash trend	26460	24856	12125	1
B06	Long/short trend	26460	25503	8776	1

8. Track B's strong validation results came with weak development cases

B01 and B03 barely cleared the positive-development-return gate. They were eligible under the frozen rule; annual consistency was diagnostic only. Their validation improvement deserves investigation without pretending it was forecast by strong development performance.

Frozen Track B cases: net results in both periods

2022–2023 development

ID	Net return	CAGR estimate	Net Sharpe 365 / 252	Annual full-L1 turnover	Hourly max DD
B01	+0.34%	0.17%	0.14 / 0.12	81.69×	37.34%
B03	+1.05%	0.53%	0.15 / 0.12	27.75×	27.64%
B04	+44.11%	20.04%	0.63 / 0.52	21.31×	61.98%
B05	+12.60%	6.11%	0.35 / 0.29	11.88×	36.50%
B06	+35.86%	16.56%	0.55 / 0.46	22.88×	48.23%

2024–2025 validation: selection statistics

ID	Net return	CAGR estimate	Net Sharpe 365 / 252	Annual full-L1 turnover	Hourly max DD
B01	+120.19%	48.31%	1.62 / 1.35	77.45×	15.50%
B03	+151.30%	58.43%	1.85 / 1.53	24.13×	22.08%
B04	+73.78%	31.78%	1.14 / 0.95	22.34×	23.42%
B05	+152.00%	58.65%	1.40 / 1.17	12.48×	32.48%
B06	+164.24%	62.45%	1.24 / 1.03	24.59×	32.83%

Returns are compounded net of 20 bps transaction costs. Sharpe uses both annualization conventions, 365 / 252, on the same daily series. Turnover is annual full-L1 target change, including entry and liquidation. Hourly max DD is the positive loss magnitude.

Annual consistency was not a Track B gate

ID	2022	2023	2024	2025
B01	+8.87%	-7.84%	+48.93%	+47.85%
B03	-9.01%	+11.06%	+26.72%	+98.31%
B04	+36.70%	+5.42%	+46.66%	+18.49%
B05	-33.96%	+70.51%	+111.32%	+19.25%
B06	+8.59%	+25.11%	+56.17%	+69.20%

9. Nine exact identities were frozen for evaluation

BTC and ETH are fixed in every basket; the table lists the other six slots. h is trailing lookback hours, r is rebalance interval hours, and b025 is the whole-target full-L1 buffer of 0.25. Weekly r168 decisions occur Monday at 00:00 UTC; r072 uses the established elapsed-time schedule.

Track A remains the original development-selected procedure; Track B remains the development-plus-validation selection. The shortlist is **not a portfolio**, has no ensemble weights, and does not designate a winner.

Case–basket identity register

Track A: four original selections

ID	Strategy	Basket	Other six coins
A01	buffered_momentum_h0672_r168_b025	4645	SOL, LUNA, FTM, TRX, LINK, LTC
A02	momentum_h0672_r072	4645	SOL, LUNA, FTM, TRX, LINK, LTC
A03	reversal_h1440_r168	1159	SHIB, DOT, ADA, MANA, FIL, VET
A04	trend_long_short_h0672_r024	4444	SOL, LUNA, AVAX, FTM, FIL, VET

Track B: five validation-informed selections

ID	Strategy	Basket	Other six coins
B01	buffered_momentum_h0672_r024_b025	2323	BNB, XRP, DOGE, FTM, AXS, VET
B03	momentum_h2160_r072	2457	BNB, XRP, FTM, MATIC, AXS, LTC
B04	reversal_h2160_r168	6064	DOT, LUNA, AVAX, MANA, FIL, VET
B05	trend_long_cash_h0672_r168	2330	BNB, XRP, DOGE, TRX, GALA, LINK
B06	trend_long_short_h0672_r168	2495	BNB, XRP, TRX, MANA, GALA, LINK

10. What was frozen, and what these results can support

Post-evaluation scope amendment. The current publication retains **168 definitions across five families**. This scope was narrowed after the development, validation and 2026 evaluations; it is not a new pre-evaluation freeze. The nine frozen IDs, baskets and settings, and the original development-ranked group memberships, are unchanged: there is **no reranking or reselection**. Population summaries are filtered or recomputed from saved retained-case results, not new strategy runs. This change supplies **no new independent OOS evidence**.

Do not move windows after inspecting performance. Development and validation each span two complete years, with no result-driven change of boundaries. Annual slices inherit the continuous path's positions and fees; there is no invented internal-year liquidation or cash reset.

Report the right Sharpe. Development Sharpe describes fitted evidence; Track A validation Sharpe describes transfer of a development-selected rule; Track B validation Sharpe is selection-biased. The 2026 evaluation must retain this selection history; later explanatory diagnostics are not a new independent test.

Separate statistical aims. Track A asks whether a development-selection procedure generalizes. Track B asks whether a strategy selected using development and validation transfers to a later period. Neither requires the development winner to remain rank one, but economic and risk expectations must be stated before that later evaluation.

Reserved evaluation protocol. The nine identities and approved risk limits remain fixed. The later approved amendment uses 1 January–30 September 2026 (273 days), rather than a complete calendar year. It requires daily-mean t-statistics, block-bootstrap intervals, multiple-testing correction and honest partial-year labels. Only the cross-period opening uses 2026 follow-up outcomes for orientation; the locked validation tables remain 2024-2025.

Common measurement and limitations

Data source and complete study dataset. Binance public Data Vision (**data.binance.vision**) spot-USDT **1-hour klines** supply prices and quote volumes. The normalized UTC grid has **43,824 hourly close endpoints**, from **1 October 2021 01:00 to 1 October 2026 00:00**, including warm-up. The packaged **22-slot screening universe** includes SAND; the nine frozen strategies use the **21-slot union of the full universe except SAND**. Lifecycle and missingness masks remain explicit. Data documentation | Full coin inventory | File hashes and coverage.

Net return compounds the saved hourly gross P&L less one-way fee × full-L1 target change, including actual entry and terminal liquidation. Annual turnover is 365 × mean daily full-L1 target change, **not** round-trip turnover or drift-adjusted executed notional. Maximum drawdown is positive peak-to-trough loss magnitude; annualized daily Sharpe uses sample SD and both 365- and 252-day conventions on the same complete UTC daily series.

The sensitivity set is 0/5/10/15/40 bps around the 20 bps primary scenario, plus 20 bps with one additional execution hour. Actual borrowing, funding, margin, order-book depth, impact and ordinary drift-maintenance trading were not modeled. Spot shorts are theoretical; historical liquidity and surviving economic slots do not remove universe or lifecycle bias.

Performance tables and fixed-basket attribution describe the same frozen strategies. Source menus expose each component's exact evidence. Detailed configurations, failed cases, cost scenarios, annual risk tables, model logs and provenance are supporting results, not additional reports.

Per-coin attribution: what each coin contributed, and what joint basket exposure explains

1 January 2024-31 December 2025, all nine unchanged frozen strategies. These explanatory diagnostics separate accounting contribution from statistical sensitivity. Strategy returns, Sharpe, turnover, drawdowns, baskets and decisions remain fixed. Track B validation alpha/beta and attribution remain selection-conditioned diagnostics; they are not unbiased final-test results.

The overview shows **annual alpha in percent per year**, estimate [95% CI]. Alpha is 365 times the daily arithmetic intercept, **not CAGR**. The BTC-only comparison uses its full-calendar HAC7 estimate; matched BTC and the joint model use the identical complete-case dates with calendar HAC27.

2024-2025 validation: all nine frozen strategies

ID	Joint alpha / yr [95% CI]	BTC alpha / yr [95% CI]	Matched BTC alpha / yr [95% CI]	Fit / calendar days	R^2	Max VIF
A01	-8.24 [-37.08, 20.60]	-6.56 [-42.44, 29.33]	-7.07 [-40.32, 26.18]	727/731	0.236	3.87
A02	-16.12 [-51.57, 19.32]	-15.33 [-54.61, 23.95]	-15.70 [-53.27, 21.87]	727/731	0.208	3.87
A03	4.47 [-17.92, 26.86]	-8.40 [-39.49, 22.69]	-8.40 [-36.24, 19.44]	731/731	0.282	5.58
A04	-1.65 [-85.25, 81.96]	2.01 [-75.00, 79.02]	4.59 [-74.54, 83.71]	727/731	0.127	3.72
B01	19.03 [-14.44, 52.50]	46.71 [10.96, 82.45]	45.14 [9.41, 80.87]	727/731	0.176	4.25
B03	11.95 [-17.16, 41.07]	52.23 [19.59, 84.87]	48.75 [17.38, 80.11]	723/731	0.223	4.34
B04	39.98 [11.80, 68.17]	31.58 [-4.94, 68.10]	31.58 [-4.17, 67.34]	731/731	0.308	5.37
B05	31.81 [1.36, 62.25]	31.35 [-18.03, 80.72]	31.35 [-20.51, 83.20]	731/731	0.634	3.96
B06	70.23 [8.68, 131.77]	70.06 [0.22, 139.89]	70.06 [3.93, 136.19]	731/731	0.101	3.83

How to read this. The joint regression includes an intercept and all eight basket coins simultaneously. A conditional beta is sensitivity holding the other seven factors fixed; it is not a portfolio weight, a causal contribution or proof of market neutrality. R-squared is an in-sample descriptive fit. Comparing joint with matched BTC separates added factors from sample exclusions; the full-calendar BTC-only comparison also uses a different covariance convention.

Uncertainty and missingness. Joint and matched BTC intervals are approximate marginal 95% normal-Wald intervals using Bartlett calendar HAC27 and n/(n-k) correction; HAC7 sensitivity is exported. Missing factor days are excluded from regressions only; gaps are not filled or compressed. The complete strategy and attribution calendars remain intact. Standardized condition >30 or any VIF >10 is a warning, not a reason to drop a coin. Some fits are sensitive to extreme days; high leverage/Cook's distance can coexist with modest collinearity. No event is removed and no model reselected.

Per-coin accounting. Gross contributions and positive allocated fees are both linked with prior **net** strategy NAV; net = gross - fees. Units are whole-period percentage points (pp) of initial capital, not standalone coin returns. The next pages retain all eight slots and their marginal net-contribution intervals.

A01 | basket 4645 | 2024-2025 validation

Buffered momentum; 28.0d lookback, 168h rebalance. **Net period return: -29.55%.** The eight net contributions below sum to that unchanged return.

Joint fit **727/731 days**; R-squared **0.236**; standardized condition **5.04**; max VIF **3.87**. Neither declared collinearity warning threshold was crossed. Max leverage **0.746**, max Cook's distance **12.496**: low collinearity alone does not establish stability. Excluded: **2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.123 [-0.185, -0.060]**; matched-calendar HAC27 **-0.123 [-0.196, -0.050]**. All brackets are marginal 95% intervals.

A01: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-0.17	1.23	-1.40	[-12.59, 11.10]	0.014 [-0.087, 0.115]
ETH	-1.80	1.22	-3.02	[-21.45, 23.47]	-0.017 [-0.119, 0.085]
SOL	0.50	1.30	-0.80	[-26.60, 31.17]	-0.005 [-0.096, 0.086]
LUNA/LUNC	-23.94	1.52	-25.46	[-66.13, 11.32]	-0.112 [-0.163, -0.061]
FTM/S	16.72	1.23	15.49	[-26.38, 78.63]	-0.021 [-0.088, 0.047]
TRX	-5.05	1.55	-6.60	[-39.58, 12.80]	-0.092 [-0.157, -0.026]
LINK	7.92	1.30	6.63	[-17.41, 31.79]	0.038 [-0.089, 0.165]
LTC	-13.02	1.36	-14.38	[-32.66, -0.54]	0.019 [-0.043, 0.081]

A02 | basket 4645 | 2024-2025 validation

Momentum; 28.0d lookback, 72h rebalance. **Net period return: -39.63%.** The eight net contributions below sum to that unchanged return.

Joint fit **727/731 days**; R-squared **0.208**; standardized condition **5.04**; max VIF **3.87**. Neither declared collinearity warning threshold was crossed. Max leverage **0.746**, max Cook's distance **13.712**: low collinearity alone does not establish stability. Excluded: **2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.100 [-0.161, -0.039]**; matched-calendar HAC27 **-0.100 [-0.176, -0.024]**. All brackets are marginal 95% intervals.

A02: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	4.58	2.00	2.58	[-8.72, 15.18]	0.064 [-0.044, 0.172]
ETH	-3.50	2.00	-5.50	[-22.54, 19.88]	-0.039 [-0.143, 0.065]
SOL	-2.07	2.24	-4.31	[-27.95, 24.00]	-0.027 [-0.111, 0.056]
LUNA/LUNC	-17.19	2.32	-19.51	[-65.67, 21.11]	-0.101 [-0.147, -0.054]
FTM/S	10.80	2.10	8.69	[-29.83, 61.43]	-0.012 [-0.085, 0.060]
TRX	-4.62	2.62	-7.23	[-37.82, 13.52]	-0.094 [-0.164, -0.024]
LINK	9.81	2.24	7.58	[-14.91, 34.85]	0.033 [-0.097, 0.163]
LTC	-19.65	2.28	-21.93	[-41.22, -4.61]	0.032 [-0.035, 0.099]

A03 | basket 1159 | 2024-2025 validation

Reversal; 60.0d lookback, 168h rebalance. **Net period return: -22.31%.** The eight net contributions below sum to that unchanged return.

Joint fit **731/731 days**; R-squared **0.282**; standardized condition **6.44**; max VIF **5.58**. Neither declared collinearity warning threshold was crossed. Max leverage **0.499**, max Cook's distance **2.425**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.026 [-0.083, 0.031]**; matched-calendar HAC27 **-0.026 [-0.089, 0.037]**. All brackets are marginal 95% intervals.

A03: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-2.26	1.04	-3.30	[-18.88, 12.96]	-0.102 [-0.174, -0.029]
ETH	3.51	0.85	2.66	[-23.48, 31.37]	0.010 [-0.085, 0.106]
SHIB	-10.60	0.98	-11.58	[-43.28, 19.02]	-0.025 [-0.078, 0.028]
DOT	13.58	1.06	12.52	[-9.85, 40.20]	0.075 [0.009, 0.140]
ADA	-22.43	1.04	-23.46	[-64.17, 12.61]	-0.114 [-0.151, -0.076]
MANA	-12.17	1.10	-13.27	[-38.20, 8.58]	0.051 [-0.004, 0.105]
FIL	11.18	1.18	10.00	[-28.22, 71.36]	0.100 [0.038, 0.163]
VET	5.25	1.13	4.12	[-20.67, 31.52]	-0.015 [-0.075, 0.046]

A04 | basket 4444 | 2024-2025 validation

Trend long/short; 28.0d lookback, 24h rebalance. **Net period return: -45.31%.** The eight net contributions below sum to that unchanged return.

Joint fit **727/731 days**; R-squared **0.127**; standardized condition **5.28**; max VIF **3.72**. Neither declared collinearity warning threshold was crossed. Max leverage **0.489**, max Cook's distance **1.315**: low collinearity alone does not establish stability. Excluded: **2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.309 [-0.548, -0.071]**; matched-calendar HAC27 **-0.304 [-0.639, 0.031]**. All brackets are marginal 95% intervals.

A04: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-1.27	3.12	-4.40	[-14.79, 20.05]	-0.001 [-0.308, 0.307]
ETH	3.21	2.24	0.97	[-11.88, 33.41]	-0.035 [-0.265, 0.196]
SOL	5.49	1.89	3.60	[-13.32, 42.91]	-0.183 [-0.311, -0.055]
LUNA/LUNC	-17.11	2.77	-19.88	[-55.52, 11.57]	-0.035 [-0.093, 0.022]
AVAX	3.15	2.24	0.92	[-12.20, 54.04]	0.036 [-0.062, 0.133]
FTM/S	-8.48	2.61	-11.08	[-45.62, 53.22]	-0.058 [-0.133, 0.017]
FIL	-10.93	2.96	-13.89	[-64.05, 22.93]	-0.080 [-0.183, 0.023]
VET	0.20	1.74	-1.55	[-18.64, 57.38]	0.118 [-0.035, 0.270]

B01 | basket 2323 | 2024-2025 validation

Buffered momentum; 28.0d lookback, 24h rebalance. **Net period return: 120.19%.** The eight net contributions below sum to that unchanged return.

Joint fit **727/731 days**; R-squared **0.176**; standardized condition **6.02**; max VIF **4.25**. Neither declared collinearity warning threshold was crossed. Max leverage **0.272**, max Cook's distance **0.427**: low collinearity alone does not establish stability. Excluded: **2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.080 [-0.142, -0.017]**; matched-calendar HAC27 **-0.081 [-0.156, -0.005]**. All brackets are marginal 95% intervals.

B01: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	8.25	5.64	2.61	[-19.92, 44.10]	-0.003 [-0.120, 0.115]
ETH	20.94	5.15	15.79	[-16.05, 58.29]	-0.074 [-0.191, 0.043]
BNB	-6.58	5.34	-11.92	[-53.40, 14.90]	0.065 [0.006, 0.124]
XRP	33.26	4.54	28.73	[-26.13, 144.54]	0.077 [-0.014, 0.167]
DOGE	36.82	5.21	31.61	[-24.32, 185.67]	0.042 [-0.024, 0.109]
FTM/S	33.03	5.68	27.34	[-61.95, 119.82]	-0.010 [-0.079, 0.060]
AXS	11.45	5.65	5.81	[-57.38, 57.29]	-0.170 [-0.236, -0.104]
VET	25.13	4.91	20.22	[-25.96, 81.90]	0.036 [-0.032, 0.104]

B03 | basket 2457 | 2024-2025 validation

Momentum; 90.0d lookback, 72h rebalance. **Net period return: 151.30%.** The eight net contributions below sum to that unchanged return.

Joint fit **723/731 days**; R-squared **0.223**; standardized condition **5.80**; max VIF **4.34**. Neither declared collinearity warning threshold was crossed. Max leverage **0.145**, max Cook's distance **0.422**: low collinearity alone does not establish stability. Excluded: **2024-09-10 to 2024-09-13 (4 days); 2025-01-13 to 2025-01-16 (4 days)**.

BTC-only beta: full-calendar HAC7 **-0.054 [-0.117, 0.008]**; matched-calendar HAC27 **-0.057 [-0.132, 0.017]**. All brackets are marginal 95% intervals.

B03: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	11.11	1.87	9.23	[-11.50, 42.98]	0.080 [-0.008, 0.168]
ETH	-3.94	1.48	-5.42	[-54.32, 49.26]	-0.024 [-0.115, 0.068]
BNB	12.99	1.94	11.06	[-22.50, 70.77]	0.088 [-0.033, 0.210]
XRP	15.65	1.58	14.08	[-36.66, 111.18]	0.109 [0.018, 0.199]
FTM/S	81.54	1.49	80.05	[-3.97, 199.55]	0.006 [-0.058, 0.069]
MATIC/POL	28.86	1.83	27.03	[-10.41, 93.97]	-0.084 [-0.140, -0.028]
AXS	35.73	1.54	34.19	[-26.51, 99.26]	-0.141 [-0.210, -0.072]
LTC	-16.77	2.14	-18.91	[-58.41, 4.28]	-0.016 [-0.121, 0.089]

B04 | basket 6064 | 2024-2025 validation

Reversal; 90.0d lookback, 168h rebalance. **Net period return: 73.78%.** The eight net contributions below sum to that unchanged return.

Joint fit **731/731 days**; R-squared **0.308**; standardized condition **6.31**; max VIF **5.37**. Neither declared collinearity warning threshold was crossed. Max leverage **0.497**, max Cook's distance **9.072**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.007 [-0.061, 0.046]**; matched-calendar HAC27 **-0.007 [-0.069, 0.054]**. All brackets are marginal 95% intervals.

B04: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-7.23	1.09	-8.32	[-34.48, 22.61]	-0.119 [-0.187, -0.051]
ETH	31.95	1.18	30.77	[-14.00, 93.27]	-0.009 [-0.114, 0.097]
DOT	16.52	1.34	15.18	[-15.45, 54.22]	-0.006 [-0.108, 0.097]
LUNA/LUNC	-18.15	1.89	-20.04	[-81.53, 52.33]	0.095 [0.002, 0.188]
AVAX	18.98	1.93	17.05	[-32.13, 68.31]	-0.041 [-0.135, 0.054]
MANA	-12.20	1.60	-13.80	[-53.79, 40.10]	0.003 [-0.080, 0.087]
FIL	52.05	1.72	50.33	[-13.89, 178.01]	0.144 [0.052, 0.236]
VET	4.21	1.60	2.61	[-42.44, 49.42]	-0.074 [-0.125, -0.024]

B05 | basket 2330 | 2024-2025 validation

Trend long/cash; 28.0d lookback, 168h rebalance. **Net period return: 152.00%.** The eight net contributions below sum to that unchanged return.

Joint fit **731/731 days**; R-squared **0.634**; standardized condition **5.20**; max VIF **3.96**. Neither declared collinearity warning threshold was crossed. Max leverage **0.754**, max Cook's distance **0.635**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **0.462 [0.361, 0.562]**; matched-calendar HAC27 **0.462 [0.318, 0.606]**. All brackets are marginal 95% intervals.

B05: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	10.99	1.31	9.69	[-4.60, 81.36]	-0.070 [-0.217, 0.078]
ETH	21.47	1.15	20.31	[-6.46, 125.01]	-0.002 [-0.104, 0.099]
BNB	26.05	1.35	24.69	[-2.87, 114.62]	0.070 [0.003, 0.137]
XRP	29.09	1.03	28.06	[-12.53, 220.06]	0.059 [-0.011, 0.129]
DOGE	12.72	1.11	11.61	[-21.25, 200.25]	0.121 [0.077, 0.166]
TRX	19.46	1.07	18.38	[-3.18, 118.96]	0.107 [0.089, 0.124]
GALA	22.16	0.82	21.34	[-11.29, 196.47]	0.086 [0.055, 0.117]
LINK	18.77	0.85	17.91	[-13.84, 109.17]	0.047 [-0.019, 0.114]

B06 | basket 2495 | 2024-2025 validation

Trend long/short; 28.0d lookback, 168h rebalance. **Net period return: 164.24%.** The eight net contributions below sum to that unchanged return.

Joint fit **731/731 days**; R-squared **0.101**; standardized condition **5.41**; max VIF **3.83**. Neither declared collinearity warning threshold was crossed. Max leverage **0.753**, max Cook's distance **0.313**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.203 [-0.394, -0.012]**; matched-calendar HAC27 **-0.203 [-0.482, 0.076]**. All brackets are marginal 95% intervals.

B06: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	7.63	2.28	5.34	[-14.89, 61.28]	-0.229 [-0.509, 0.051]
ETH	23.99	2.03	21.96	[-11.60, 128.28]	-0.115 [-0.334, 0.104]
BNB	18.85	2.42	16.43	[-12.50, 97.53]	0.016 [-0.124, 0.155]
XRP	14.54	1.83	12.71	[-30.79, 158.68]	0.009 [-0.110, 0.127]
TRX	11.63	1.79	9.84	[-11.24, 92.69]	0.106 [0.064, 0.149]
MANA	30.65	1.54	29.11	[-13.82, 168.05]	0.070 [-0.057, 0.197]
GALA	53.80	1.36	52.43	[0.34, 259.05]	0.046 [-0.053, 0.145]
LINK	17.95	1.53	16.41	[-29.09, 104.59]	-0.043 [-0.173, 0.086]

Full precision, sensitivity and multiplicity

Per-coin gross, fee and net attribution with all three confidence intervals | Joint alpha/betas, HAC7 sensitivity, global Holm243 p-values and Bonferroni243 intervals | Joint model diagnostics and individual VIFs | Full-calendar BTC-only estimates and intervals | Matched BTC-only coefficients and intervals | Regression calendar and exclusions.

Attribution uses **4,000 seeded, synchronized circular 28-day calendar-block draws** across all nine strategies and eight coins. Daily gross/fee vectors are resampled together and relinked using each draw's **new prior NAV**; signals, targets and trades are not regenerated and no block-boundary fee is invented. Any invalid draw makes the affected interval unavailable; none is silently dropped. Gross/fee intervals and invalid-draw counts remain in the CSV.

The coefficient correction spans **243 predeclared diagnostic slots: 27 joint alphas + 216 joint betas** across all three periods. Displayed intervals are marginal; approximate Bonferroni familywise intervals and Holm-adjusted p-values are exported. This corrects only that conditional coefficient family, **not the original universe, configuration, period or strategy-selection search**. Attribution intervals are marginal and not multiplicity-adjusted. No significance row changes the frozen shortlist or crowns a new winner. LUNA/LUNC is legacy Terra, not Terra 2.0; FTM/S and MATIC/POL retain the original economic slots.